The Alberta Buck - Demurrage and the Jubilee Fund

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Abstract

Held BUCK pay demurrage on the BUCK remaining in the account. Each BuckCredit position earns Jubilee relief on the BUCK it actually issued, proportional to issuance and elapsed time. The total gross demurrage basis is all outstanding BUCK, including BUCK held by Jubilee. The fund receives the charges and may invest its assets; it pays the relief when issuance is retired. Gross demurrage includes charges paid by Jubilee to itself, so cumulative gross demurrage and BUCK actually held by the fund are distinct.

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1 Status

This is the intended monetary design, clarified on 2026-10-27. The executable Python reference model and its tests exercise the mathematics below. The current Solidity contract and Rust mirror still use a linear charge on a holder's gross raw balance and advance Jubilee on issuance; they do not yet implement the model in this document. Their existing tests establish the old behavior, not conformance with the new one. The detailed implementation audit is Net-balance demurrage: audit and redesign.

Earlier versions of this document described linear demurrage on the holder's raw balance and contained older storage sketches. Those formulas are superseded; the previous text remains available in the Git history.

2 Issuance, holdings, and relief

A BuckCredit authorizes credit; only the BUCK actually issued into existence forms the demurrage base and earns relief. Undrawn credit is not itself BUCK. While issuance remains outstanding, all of those BUCK are held somewhere: ordinary accounts, pool or other service accounts, or Jubilee. The underlying asset's insured coverage and a user's unused credit headroom are different quantities from the number of BUCK actually issued.

Let `I(t)` be total outstanding BUCK issuance, `r` the nominal annual rate, and `P_j(t)` the outstanding issuance attributable to credit position `j`. As issuance changes, the system checkpoints the old base and starts integrating the new base. Over an interval,

```
BUCK-years issued       = integral(I(t) dt)
gross demurrage         = r * integral(I(t) dt)
position j BUCK-years   = integral(P_j(t) dt)
position j target relief = r * integral(P_j(t) dt)
```

BUCK-years are the integral of outstanding issued BUCK, not merely the age of an open but unused credit limit. When an issuance position is reduced, its target relief is computed from its BUCK-years. Jubilee first redeems a proportion of its investment positions and computes the actual relief from its available full-value BUCK; the account repays the remaining principal. Actual relief can be lower or higher than the target, but is bounded by the principal being reduced and by the available net BUCK. In the reference, a partial reduction takes a pro-rata share of that position's accumulated BUCK-years. Separate positions retain separate issuance histories. For issuance held for 50 years at a nominal 2% per year, the target reaches the entire position; the actual payment depends on the fund and cannot exceed the position.

An account's actual BUCK holding, the outstanding issuance it owes, and its unused credit are not interchangeable. Relief reduces a liability only when it is paid. The target and the actual relief computed after liquidation must each be visible; a target shortfall is not an additional guaranteed claim.

3 Demurrage on remaining balances

Demurrage already transferred to Jubilee no longer belongs to the former holder and is not charged again to that holder. For an idle account with a remaining balance `H` and no other flows, the Python reference uses a nominal continuous rate:

```
dH/dt = -r H
H(t)  = H(0) * exp(-r t)
```

The reference uses continuous compounding because the contract has a per-second nominal rate and a result must not depend on the frequency of permissionless settlement. If a future implementation instead chooses an effective annual rate or another schedule, that convention and its rounding must be specified and used consistently in contracts, models, and tools.

New incoming BUCK join the remaining balance at their arrival time. Outgoing BUCK leave the balance at their outgoing time. Subsequent charges apply to the resulting net holdings, never to fees already transferred out. For example, an account with 0.5 BUCK remaining incurs approximately 0.009901 BUCK of continuous demurrage over the next year at 2%; it does not pay 2% of a much larger historical raw balance.

The fund's BUCK also experience demurrage. If the fund holds BUCK directly, its demurrage is transferred to the fund itself: it counts in total gross demurrage, but the self-transfer has no net effect on fund assets. A fund position in a basket or desk requires separate accounting for the underlying BUCK, beneficial ownership, realizable asset value, and its returns. It cannot assume that a mark or a cumulative demurrage figure is immediately spendable BUCK.

4 The 100 BUCK, 50-year example

One credit issues 100 BUCK to an ordinary wallet. Nothing is transferred, invested, retired, or added for 50 years. At a nominal continuous 2% rate, the Python reference produces, ignoring rounding:

Years	Wallet balance	Jubilee BUCK	Gross demurrage	Jubilee self-demurrage	Target relief
0	100.000000	0.000000	0.000000	0.000000	0.000000
1	98.019867	1.980133	2.000000	0.019867	2.000000
25	60.653066	39.346934	50.000000	10.653066	50.000000
49	37.531110	62.468890	98.000000	35.531110	98.000000
50	36.787944	63.212056	100.000000	36.787944	100.000000

The 100 BUCK gross demurrage is the sum of 63.212056 transferred from the wallet and 36.787944 paid by Jubilee to itself. The latter is counted as a charge, but does not create an additional 36.787944 BUCK in the fund. Wallet plus fund still hold 100 BUCK, matching the outstanding issuance. Crediting Jubilee with all 100 **in addition** to leaving 36.787944 in the wallet would create 36.787944 BUCK without a source.

The target relief at year 50 is 100, but the fund directly holds only 63.212056 in this no-return, no-transfers example. It needs another 36.787944 BUCK to pay the full target. Acquiring that amount from the wallet, for example by selling other assets or realizing returns, would leave 100 in the fund; a 100 BUCK Jubilee payment can then extinguish the credit. The external consideration and market price are not represented by the closed BUCK ledger. Under the available-relief rule the actual relief is computed from the fund's net liquid BUCK after any proportional redemption; the borrower pays the remainder. No separate fixed 100 BUCK payout is assumed.

5 Fund investments and liquidity

Jubilee may put its assets to work in the BUCK-maximizing BuckBasket and, when available, desk implementations that seek other assets such as USDC. A useful accounting implementation must

distinguish:

- cumulative gross demurrage, including Jubilee's self-payments;
- liquid BUCK owned by the fund;
- deployed positions, their acquisition cost, current mark and realizable proceeds after liquidity, price impact and transaction costs;
- other assets and the BUCK they can actually acquire;
- each credit position's accrued and payable relief; and
- the cash and positions reserved for anticipated relief payments.

For the fund's cash ledger, separate cumulative 256-bit counters can record full-value demurrage received, relief paid, BUCK invested, and net BUCK withdrawn. With no other flows, liquid net BUCK equals received demurrage plus investment withdrawals, less relief payments and investments. Other inflows and outflows are recorded separately. Demurrage paid by the fund to itself cannot be added to net incoming cash a second time. A running average cost basis on the pooled investment additionally identifies gains or losses only as the corresponding shares are withdrawn. The net of either pair of cumulative counters can be negative after gains; unsigned totals must not be subtracted without signed or otherwise guarded arithmetic.

When a BuckCredit position is reduced, the fund first redeems a proportional slice of its positions, then computes relief from the net BUCK in the fund. Even a redemption returning zero BUCK leaves a well-defined available relief, possibly zero. The borrower supplies the difference; if the borrower lacks that amount, the reduction cannot complete. A modeled investment return must have a source, such as a transfer from a counterparty, a realized external asset sale, or new, separately recorded issuance. It must not appear solely because the fund paid demurrage to itself. The reference includes an idealized BUCK-only strategy that pays demurrage into fund cash and liquidates at its current mark, plus returns transferred explicitly from a counterparty. The strategy's mark falls as its BUCK pay fees, while the same fee increases fund cash: counting the mark decrease alone as a loss of total fund value would be misleading. The reference does not model actual basket prices, collateral values, transaction costs, delay, or market depth.

The reference's available-relief experiment uses a reserve target of 10% of the remaining baseline relief and scales the target relief by the ratio of post-redemption net cash to that reserve. It caps the result at the net cash and the principal. This is a model of the proposed reserve signal, not a final contract rule: a redemption itself changes cash, so the signal, the redemption fraction, and their effect on payout order require comparison. The simulator permits the fraction to be supplied explicitly. A fixed-2% quote and retirement are retained only as comparison diagnostics. The command-line reference defaults to the requested target relief's share of total outstanding target relief and can also compare the share of total issued BUCK. Credits of different ages can make these two fractions very different. The reference sums its positions to obtain this benchmark; an efficient contract implementation of the same aggregate remains to be designed.

6 Signed accounts and carrying services

The present contract uses one signed raw account balance and a timestamped `BUCK-seconds` accumulator. Its positive side represents gross held BUCK and currently uses the accumulator for

the holder's accrued fee; its negative side represents actual credit drawn and uses the accumulator to integrate the outstanding lien for relief. Zero must have zero accumulated seconds. A sign change must settle the old interpretation before recording the new one. A negative raw balance is an actual liability, not unused credit and not an exhausted positive holding. The signed view of an ordinary positive holding must never turn negative merely because its accumulated fee exceeds its raw amount.

The existing Carrying mode lets service contracts, including AMM pools, report nominal raw balances to preserve ERC-20 reserve accounting, while passing a proportional accumulated fee with a transfer. These nominal amounts are not necessarily the same as unencumbered economic value. The corrected model must preserve a clear distinction between nominal pool balances, remaining value, transferred fee attribution, and the fund's actual assets. For Jubilee's own Carrying balance, a relief payment must deliver its full net value. If the fund has 100 nominal BUCK with 20 BUCK of carried fee, its net balance is 80. To deliver 40 net BUCK under proportional continuous fee allocation, it sends 50 nominal BUCK carrying 10 of fee; 50 nominal and 10 of fee remain, or 40 net. The fund's already-credited self-demurrage must not be credited again on payout. The actual on-chain rounding must follow the chosen accumulator representation; the Python integer example is not yet a derivation for Solidity's buck-seconds rounding. The current sign encoding and Carrying rules are implementation details to validate or replace; they do not justify charging an already transferred fee again.

Delegated demurrage also needs an explicit net-balance treatment: its payer must have enough actual remaining value, and the outcome must remain conserved when a payer is depleted or multiple sponsees settle in different orders. See the implementation audit for the existing lazy behavior.

7 Run and verify

```
make nix-venv-demurrage-reference
make nix-venv-demurrage-reference DEMURRAGE_ARGS="--years 50 --json"
make nix-venv-demurrage-reference DEMURRAGE_ARGS="--years 50 --acquire-all"
make nix-venv-demurrage-reference DEMURRAGE_ARGS="--deploy-fund --strategy-return 10"
make nix-venv-demurrage-reference DEMURRAGE_ARGS="--deploy-fund --wipe-strategy --retire 50 --re
make nix-venv-demurrage-test
```

The reference is an event-driven, floating-point monetary model, not an EVM or fixed-point implementation. Its tests cover 1 to 1,000-year idle holdings, changes to outstanding issuance, proportional reductions, separate positions, small net balances, settlement-frequency independence, fund self-demurrage, strategy returns and liquidation, zero proceeds, grossed-up Carrying relief, conservation, and insufficient debtor funds. Before contract deployment, implement and cross-check fixed-point precision and rounding, supply and claim conservation, settlement and retirement transitions, pool and delegated-fee semantics, fund liquidity, and migration of any existing account history. The five-year economy and sandboxes must subsequently use the implemented rules; existing runs use the old contract and do not validate this design.